



Cyber Security Project: Splunk Ignite

Project Title: Log Analysis & Threat Detection Using Splunk: Data
Ingestion to Dashboarding

By: Joshua Renzo Gofulco
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1. Project Description

This project focuses on ingesting and analyzing login event data using Splunk’s Search & Reporting capabilities. Learners upload a structured dataset, apply SPL queries to identify login success and failure patterns, and build interactive dashboards to visualize trends. The goal is to develop practical skills in log analysis, data-driven investigation, and dashboard creation within a cybersecurity context.

1.1 Project Objectives

- Understand how to ingest and structure log data within Splunk.
Learn to write SPL queries for analyzing login success and failure events.
- Develop interactive dashboards to visualize authentication trends and detect anomalies.

2. Dataset Preparation

Dataset Used:

- **Source:** sample_logins.csv (Self-Created)
- **File Type:** CSV
- **Fields:** timestamp, username, ip_address, status (success/failure)

Figure 1: Sample_Login.csv Dataset

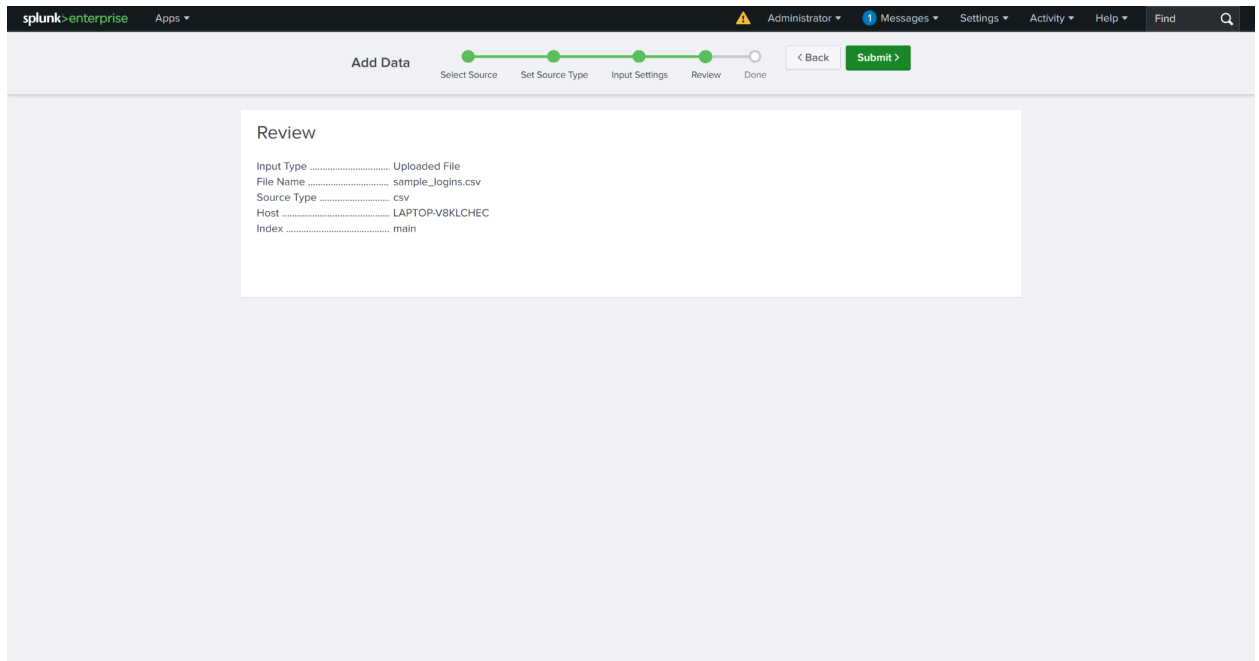
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3. Data Upload to Splunk

Steps Taken:

1. Opened Splunk → **Search & Reporting**.
2. Navigated to **Settings** → **Add Data** → **Upload**.
3. Selected dataset file (sample_logins.csv).
4. Choose source type: CSV.
5. Assigned to index: main
6. Submitted and confirmed upload.

Figure 2: Screenshot of Data Upload in Splunk



4. SPL Queries and Analysis

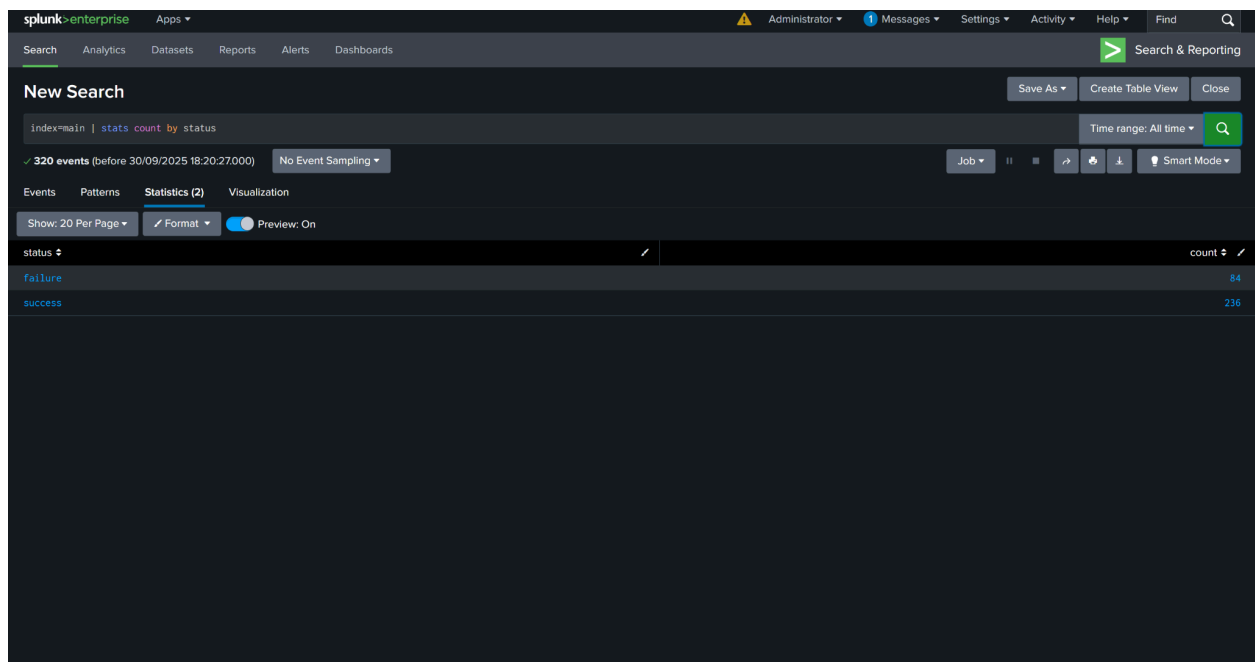
4.1 Query 1: Count of Login Success vs Failure

“index=main | stats count by status”

Explanation:

This query counts the total number of successful and failed login attempts to identify overall authentication trends.

Figure 3: Count of Login Success vs Failure Query Result



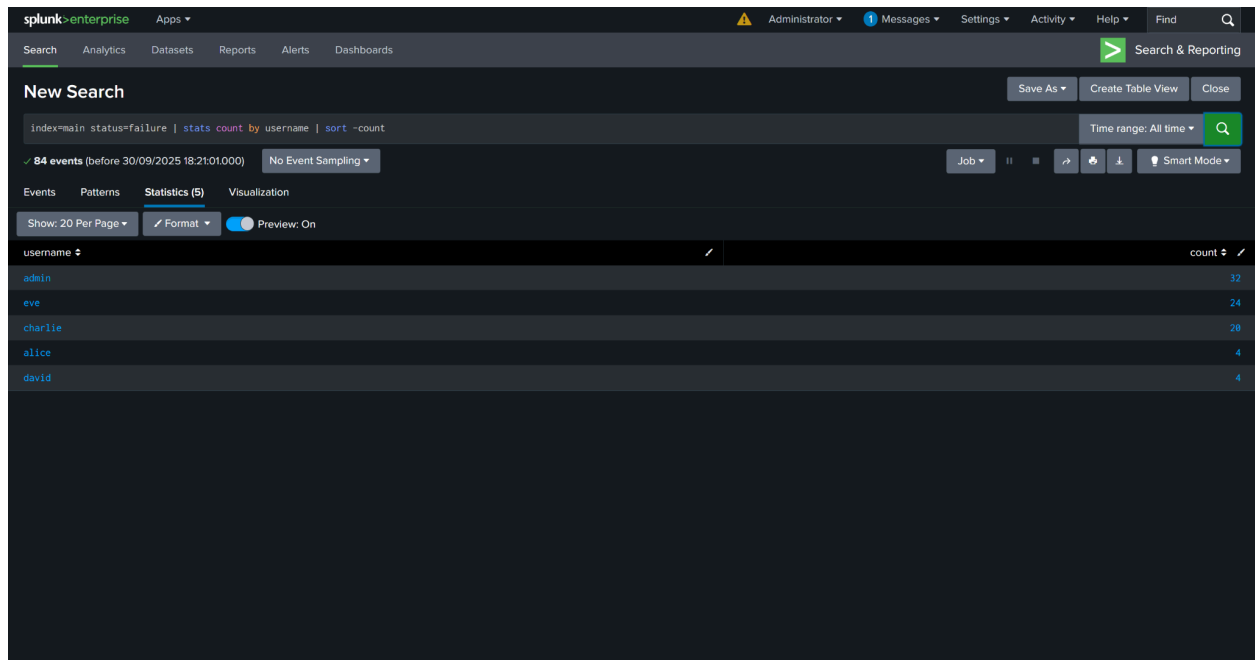
4.2 Query 2: Top Users by Failed Logins

“index=main status=failure | stats count by username | sort -count”

Explanation:

Identifies accounts with frequent failed login attempts, which may indicate brute-force attacks or compromised credentials.

Figure 4: Top Users by Failed Logins Query Result



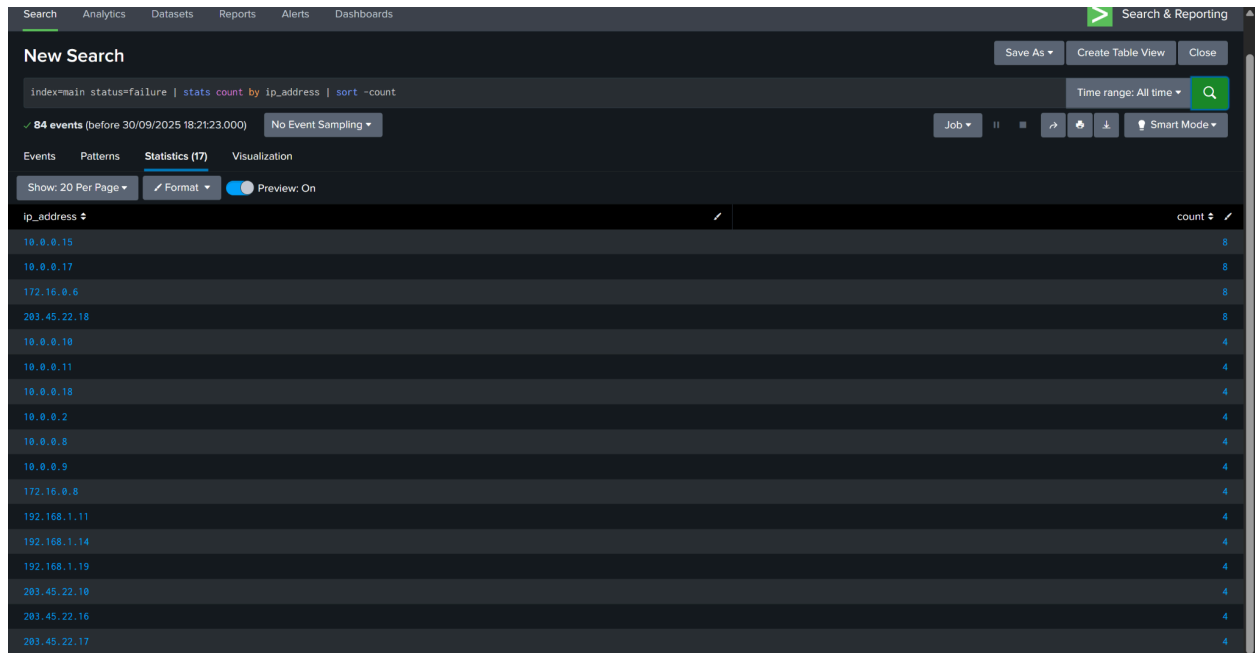
4.3 Query 3: Top IP Addresses with Failed Logins

“index=main status=failure | stats count by ip_address | sort -count”

Explanation:

Detects potentially malicious IP addresses responsible for repeated failed login attempts.

Figure 5: Top IP Addresses Query Result



The screenshot displays the Splunk search interface. At the top, there's a navigation bar with tabs for Search, Analytics, Datasets, Reports, Alerts, and Dashboards. The 'Search' tab is active. Below the navigation bar, the 'New Search' section shows the query: `index=main status=failure | stats count by ip_address | sort -count`. The search results are displayed in a table with two columns: `ip_address` and `count`. The table shows the top 20 IP addresses with the highest number of failed login attempts. The first three IP addresses have a count of 8, while the others have counts of 4 or 1.

ip_address	count
10.0.0.15	8
10.0.0.17	8
172.16.0.6	8
203.45.22.18	8
10.0.0.10	4
10.0.0.11	4
10.0.0.18	4
10.0.0.2	4
10.0.0.8	4
10.0.0.9	4
172.16.0.8	4
192.168.1.11	4
192.168.1.14	4
192.168.1.19	4
203.45.22.10	4
203.45.22.16	4
203.45.22.17	4

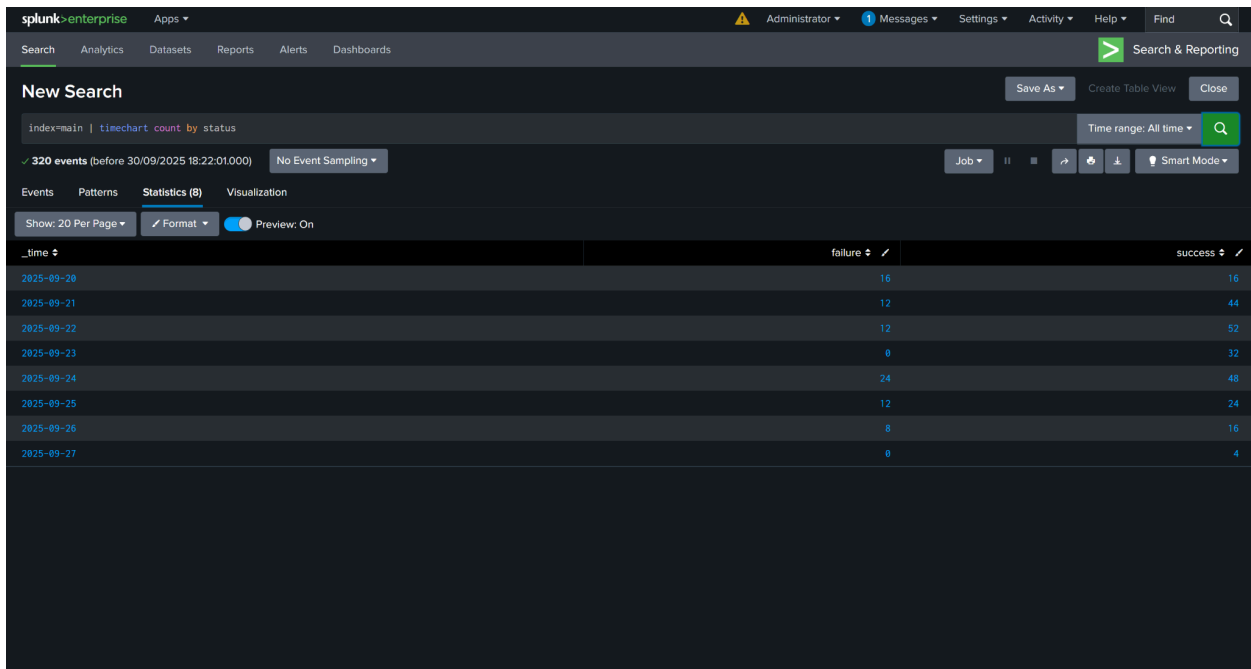
4.4 Query 4: Login Trends Over Time

“index=main | timechart count by status”

Explanation:

Shows login activity trends over time and helps identify unusual spikes or patterns outside normal working hours.

Figure 6: Login Trends Over Time Query Result



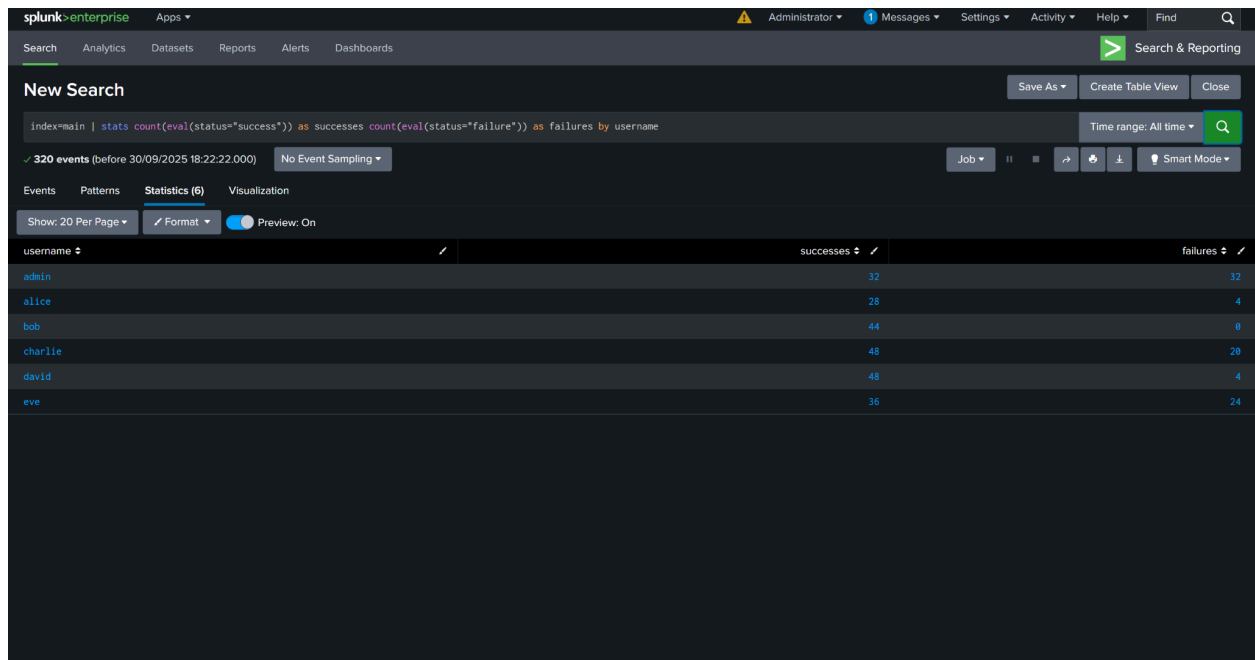
4.5 Query 5: Success-to-Failure Ratio per User

“index=main | stats count(eval(status="success")) as successes count(eval(status="failure")) as failures by username”

Explanation:

Highlights users with suspicious login behavior (e.g., many failures before a successful login), useful for detecting password spraying or compromised accounts.

Figure 7: Success-to-Failure per User Query Result



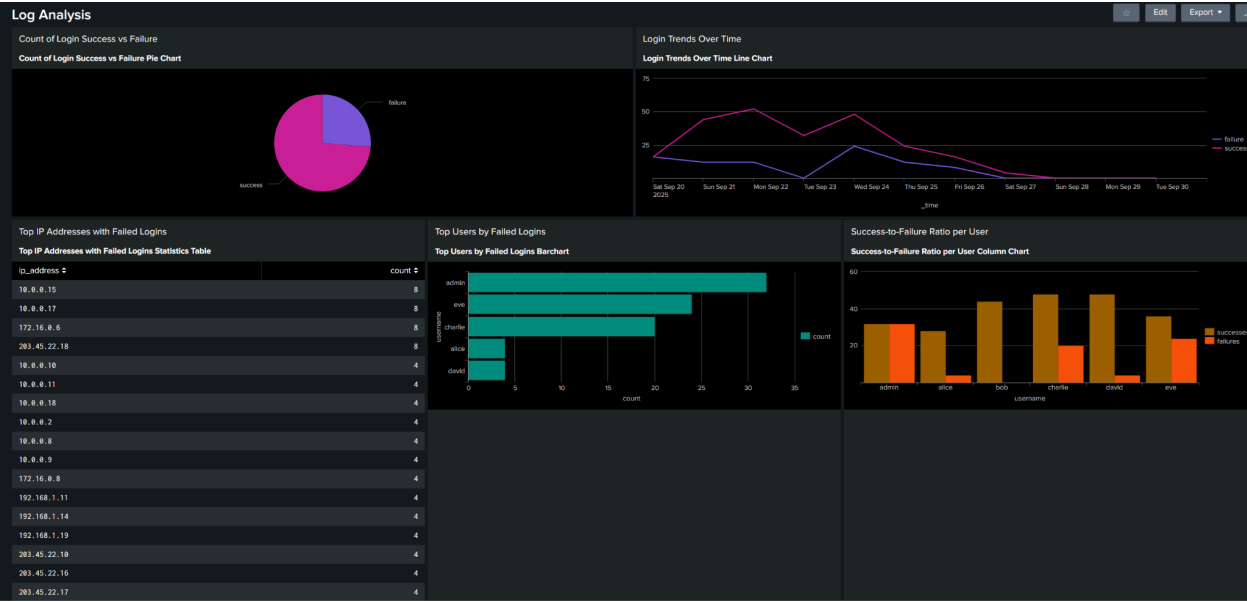
5. Dashboard Creation

Dashboard Name: Login Analysis

Panels Included:

1. **Login Success vs Failure (Pie Chart)**
2. **Top Users by Failed Logins (Bar Chart)**
3. **Top IPs by Failed Logins (Statistics Table)**
4. **Login Trends Over Time (Line Chart)**
5. **Success-to-Failure Ratio per User (Column Chart)**

Figure 8: Log Analysis Main Dashboard



5.1 Observations from Dashboard

**The situations below are explicitly used in this scenario (e.g. external IP's identified, early morning attacks, etc.)*

1. Login Status Distribution (Pie Chart)

- The chart highlighted that while the majority of login attempts were successful, there was a noticeable number of failures.
- This suggests either repeated user input errors or possible attempts at unauthorized access.

2. ***Top Users by Login Failures (Bar Chart)***

- Certain users appeared consistently with high failure counts.
- This may indicate weak passwords, forgotten credentials, or that these accounts are being actively targeted for brute-force attempts.

3. ***Top IPs for Failed Logins (Statistics Table)***

- Several IP addresses stood out with repeated failed attempts.
- Some were external addresses not linked to known user locations, which raises concerns of malicious probing or credential-stuffing activity.

4. ***Login Trends Over Time (Line Chart)***

- Login failures spiked during unusual hours (e.g., late night or early morning).
- Such timing patterns are inconsistent with normal business operations and could signal automated attacks running during off-hours.

5. ***Success-to-Failure Ratio per User (Column Chart)***

- In multiple cases, accounts logged in successfully after multiple failures.
- This pattern could point to attackers eventually guessing correct passwords, representing a serious risk to account integrity.

6. Findings & Conclusion

6.1 Key Findings Summarized

The dashboard revealed several key findings:

- **Login Status Breakdown:** Most logins were successful, but a notable share of failures indicates possible credential issues or unauthorized attempts.
- **Top Users by Failures:** A few accounts showed repeated failed logins, suggesting brute-force attempts or compromised credentials.
- **Suspicious IPs:** Multiple failed attempts came from external IP addresses not linked to the internal network, raising red flags for malicious activity.
Time-based Anomalies: Spikes in failed logins occurred outside normal working hours, a common sign of automated attacks.
- **Success After Failures:** Some users eventually logged in successfully after many failed attempts, pointing to possible password spraying or guessing

6.2 Conclusion

The dashboard analysis uncovered critical insights into user authentication activity, including a high proportion of failed logins, repeated failures from specific accounts, suspicious external IP activity, and login attempts during abnormal hours. These findings strongly suggest risks ranging from weak credentials and user errors to targeted brute-force or credential-stuffing attacks.

This highlights that log analysis plays a critical role in cybersecurity monitoring by transforming raw event data into actionable information. By correlating login behaviors, identifying anomalies, and highlighting suspicious patterns, organizations can detect threats earlier, respond faster, and reduce the likelihood of successful intrusions. This proactive visibility helps security teams enforce stronger authentication policies, improve incident response, and maintain system integrity in the face of evolving cyber threats (Dolníček, 2025).

7. References

Dolníček, L. (2025). *What Is Log Analysis: A Complete Guide*. [online] Logmanager. Available at: <https://logmanager.com/blog/log-management/what-is-log-analysis/>.